# Samuel Stefano Teixeira do Carmo

**FULL-STACK ENGINEER · APPLIED AI · TECH LEAD**

**Email** samuelstefanodocarmo@gmail.com | **Phone** +55 44 99879-5387
**Portfolio** samuelstefano.dev | **GitHub** github.com/SamuelStefano | **LinkedIn** linkedin.com/in/samuel-stefano
**Location** Brazil (UTC−3) · Remote worldwide · Full overlap with US Eastern hours

## SUMMARY

Full-stack engineer with 3+ years shipping production TypeScript end to end, currently tech lead of DevFellowship's internal developer platform — the tooling and workflow a 5-engineer team uses to build client software. Focused on **applied AI in product**: 4 Node.js MCP servers giving LLM agents typed, JWT-scoped access to internal systems, AI code review running in CI, and a purpose-built coding-agent harness. Full-stack across the whole surface: Lesson Studio, a video authoring platform with a deterministic animation engine and server-side rendering; Module Federation micro-frontends (6 apps in production); Node/NestJS APIs; PostgreSQL with row-level security; Docker Swarm in production. 1,000+ merged pull requests in the company organization and 110+ peer code reviews. Four-time award winner at international Web3 hackathons, including 1st place at ETH LatAm 2025.

## CORE SKILLS

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|---|---|
| **LANGUAGES** | TypeScript · JavaScript · SQL · Solidity |
| **FRONTEND** | React · Next.js · Vite · Module Federation / Micro-frontends · Zustand · TailwindCSS · Storybook |
| **BACKEND** | Node.js · NestJS · REST APIs · Serverless / Edge Functions · JWT & OAuth · Webhooks · Cron & background jobs · Server-side video rendering (FFmpeg) |
| **DATA** | PostgreSQL · Supabase · Prisma · Row-Level Security · MySQL · MongoDB |
| **INFRASTRUCTURE** | Docker · Docker Swarm · Linux / VPS · Vercel · CI/CD (GitHub Actions) · AWS S3 · Observability |
| **AI ENGINEERING** | Claude & OpenAI APIs · MCP servers · LLM agents & tool use · Agent harnesses (hooks, skills, sub-agents) · Sandboxed code execution · AI code review in CI · Prompt engineering & evaluation |
| **WEB3** | Solidity · ERC-20 / ERC-721 · Base · Scroll · Solana / Anchor · Chainlink CCIP & CRE · Wagmi/Viem |
| **PRACTICES** | System design · Code review & mentoring · Testing (Playwright, Vitest) · Application security · Spec-driven development · Agile |

## EXPERIENCE

### DevFellowship CURRENT — Jul 2024 — Present

**Tech Lead — Internal Developer Platform**

*Software studio · Remote · Full-time since Feb 2026*

- Tech lead of the internal engineering workflow: the 6-app micro-frontend ecosystem (Module Federation host plus Learn, Reviews, Payments, Flows and an AI code-review bot) that developers use day to day to ship client software. Set the standards a 5-engineer team builds against; 1,000+ merged PRs and 110+ reviews of teammates' PRs.
- One of two core engineers on Lesson Studio, a platform for authoring lessons and short-form video (240+ merged PRs in the product): designed the deterministic animation engine — keyframes, formulas and a catalog of 30+ effects — with parity between the editor and the server-side render, and fixed audio/video sync and audio-clipping defects in the export pipeline at the root cause.
- Built 4 Node.js MCP servers that give LLM agents typed, permission-scoped access to internal work, learning, billing and planning data via delegated JWT auth, replacing manual ops requests with self-service agent workflows.
- Put AI code review in CI and introduced a multi-agent review squad plus a spec-driven development workflow, now the team default for money- and security-critical features — design decisions are written and challenged before implementation.
- Designed and shipped the billing pipeline the company runs its revenue through — contracts → recurring invoicing → Brazilian e-invoices → instant Pix payments — with idempotent cron dispatchers, automatic recovery from stuck states, and soft-launch gates that let each stage roll out without downtime.
- Rolled out centralized secret management (Infisical) and own production operations: Docker Swarm on bare VPS, GitHub Actions pipelines, Postgres row-level security and security review of every client-facing surface.
- Represented the company at four international Web3 hackathons, shipping a demoable product in 48–72 hours each time and placing in all four (see Awards).

**STACK:** TypeScript · React · Next.js · Module Federation · Node.js · NestJS · Supabase · PostgreSQL · FFmpeg · Docker Swarm · Claude API · MCP · GitHub Actions

### M&M Informática

**Dec 2024 — Jan 2026**

**Technical Support Analyst** (full-time employee)

*ERP and retail software vendor · Hybrid*

- Supported the company's ERP customers, diagnosing system, database and store-operation incidents through to resolution.
- Worked daily against the ERP's Firebird database, investigating data and queries to close support tickets; trained on the legacy product and built working familiarity with how a retail ERP fits together — tax invoicing, point of sale and data modeling.

**STACK:** ERP support · Firebird · SQL · Incident diagnosis

### Instituto Educar+ CURRENT

**Jan 2025 — Present**

**Programming Instructor** (part-time, non-profit)

*Free vocational tech program for teenagers from underserved communities*

- Teach programming fundamentals, web development and engineering practices to classes of students with no prior technical background.
- Designed a "zero to deployed product" curriculum on Vercel and Supabase; mentored final projects one-on-one, with students going on to present their work at regional hackathons.

**STACK:** JavaScript · HTML · CSS · Git & GitHub · Vercel · Supabase

### Marialva City Hall

**Feb 2023 — Dec 2024**

**IT Support Intern**

- Supported networks, servers and workstations across multiple municipal departments, diagnosing incidents end to end for non-technical users.

## SELECTED PROJECTS

### MintWork — Verifiable work-credential layer · UNICEF Venture Fund 2026 finalist

A protocol that turns real delivered work into a verifiable credential, so youth without formal experience can prove what they have actually built. Role: **protocol and on-chain engineer** — 7 EAS schemas and 3 Solidity resolvers, a non-transferable ERC-721 credential token and the issuer/verifier SDK, with PII kept off-chain and hashes only on-chain. Shortlisted for the UNICEF Venture Fund 2026 (US$100k grant call); protocol core published under Apache-2.0.

*Solidity · Foundry · EAS (Ethereum Attestation Service) · Base · Scroll · Wagmi/Viem · Node.js · PostgreSQL · IPFS*

### Lesson Studio — Authoring platform for lessons and short-form video

Slide-based web editor for producing lessons and short-form video: in-browser audio, webcam and screen recording, captions, templates and MP4 export through server-side rendering. Deterministic animation engine (animation as a pure function of time, with keyframes, formulas and 30+ effects) with editor-to-render parity, a mobile version, and an MCP server that lets AI agents author, animate and export videos. In production for the company's lesson and social media content.

*React · TypeScript · Vite · Zustand · Supabase · FFmpeg · Headless Chromium · MCP · Playwright · Vitest*

### DFL Learn — Micro-frontend super-app

Module Federation host unifying LMS courses, a project Kanban, billing, meetings, a CMDB and an AI chat into one surface — independently deployable micro-apps behind a single shell, in production org-wide.

*Next.js · NestJS · Module Federation · Supabase · Prisma · TailwindCSS*

### Uber Money — On-chain credit scoring with Chainlink · github.com/SamuelStefano/UberMoney

Off-chain credit decisioning in a Chainlink CRE workflow, written cross-chain via CCIP (Ethereum → Solana). Anchor program with an Ed25519 attestation pattern that closes a signature-forgery hole.

*Anchor · Solana · Chainlink CCIP / CRE · Ethereum · Next.js*

### Deck — Control plane for AI coding agents · github.com/SamuelStefano/cockpit

Harness for AI coding agents: agent chat, live tmux terminals over WebSockets, persistent memory, reusable skills, token-cost tracking and a durable prompt queue that survives restarts and quota limits.

*React · TypeScript · Node.js · WebSockets · tmux · Supabase*

## AWARDS

**1st** **PIX On-Chain** — *Base track · ETH LatAm 2025 (international Web3 hackathon)*

**2nd** **Uber Money** — *Hackanation 2026 · Web3 / RWA · TokenNation, São Paulo*

**2nd** **TalentDAO** — *Scroll track · DevConnect ETH 2025*

**4th** **GreenLoop** — *Base track · ETH LatAm 2025*

## EDUCATION & LANGUAGES

**UNINGÁ — Centro Universitário Ingá** · Associate Degree in Systems Analysis and Development · On-campus · 2024 — Dec 2026 (final year · expected graduation Dec 2026)

**Portuguese** — Native · **English** — Professional working proficiency (B2): daily technical writing, documentation and code review.